Question	Answer	Marks	Partial Marks
1(a)(i)	120	**2**	**M1** for 6 ÷ (21 – 19) oe soi or for $\frac{2x}{40}=6$
1(a)(ii)(a)	34	**2**	**M1** for $40-\frac{15}{100}\times 40$ oe or better or **B1** for 6
1(a)(ii)(b)	35	**2**	**M1** for $\left(1-\frac{15}{100}\right)\times p = 29.75$ or better
1(b)(i)	44 274 cao	**3**	**B2** for 44273 to 44274 or 44270 or **M1** for $40100\times\left(1+\frac{2}{100}\right)^5$ oe
1(b)(ii)	2019 nfww	**3**	**M2** for one correct trial of $n = 8$ or $n = 9$ either to find a salary or, if working with 1.02^n and 47 500÷ 40 100 [= 1.1845], to find a value of 1.02^n or **B2** for final answer 9 or 4 nfww or **M1** for *their* $44\,274\times\left(1+\frac{2}{100}\right)^n = 47\,500$ oe or $40\,100\times\left(1+\frac{2}{100}\right)^n = 47\,500$ oe or for at least one trial giving a value greater than *their* 44 274
1(c)	2.9 [increase]	**2**	**M1** for $\left(1+\frac{5}{100}\right)\times\left(1-\frac{2}{100}\right)$ oe implied by 1.029 or 102.9[%]
2(a)(i)	–1	**2**	**M1** for $3\times 2^2 - 13$ oe
2(a)(ii)	$[\pm]\sqrt{\frac{y-t}{p}}$ oe final answer	**3**	**M1** for correct rearrangement to isolate x^2 term **M1** for correct division by p **M1** for correct square root Incorrect answer scores a maximum of M2 If 0 scored, SC1 for a correctly rearranged formula with $p = 3$ and $t = -13$ substituted

Question	Answer	Marks	Partial Marks
2(b)(i)	$(5x-4)(3x+2)$ oe final answer	**2**	**B1** for $(ax+b)(cx+d)$ where either $ac = 15$ <u>and</u> $bd = -8$ or $ad + bc = -2$ or $5x(3x + 2) - 4(3x + 2)$ or $3x(5x - 4) + 2(5x - 4)$ or correct factors seen and spoiled
2(b)(ii)	$\frac{4}{5}$ oe and $-\frac{2}{3}$ oe	**1**	**FT** a factorised quadratic
2(c)	$x(x+4y)(x-4y)$ final answer	**3**	**B2** for $(x^2+4xy)(x-4y)$ or $(x+4y)(x^2-4xy)$ or answer in the form $x(a + b)(a - b)$ or correct answer seen and spoiled or **B1** for $x(x^2-16y^2)$ oe or $(x+4y)(x-4y)$
2(d)	$\frac{1-2a}{x}$ oe final answer	**4**	**B2** for $(2x - 1)(1 - 2a)$ oe or **B1** for $2x - 1 - 2a(2x - 1)$ or $2x(1 - 2a) - (1 - 2a)$ **B1** for $x(2x - 1)$
3(a)(i)	4	**1**	
3(a)(ii)	7	**1**	
3(a)(iii)	8	**1**	
3(b)(i)	14	**1**	
3(b)(ii)	4	**2**	**B1** for [l.q. =] 11 or [u.q =] 15
3(c)	8.09	**3**	**M1** for $5 \times 3 + 10 \times 6 + 43 \times 7 + 75 \times 8 + 48 \times 9 + 21 \times 10$ **M1 dep** ÷ 200
3(d)	30, 70, 40, 36, 24 seen	**B2**	**B1** for 3 or 4 correct or **M1** for $1 \times (80 - 50)$, $3.5 \times (100 - 80)$, $4 \times (110 - 100)$, $3.6 \times (120 - 110)$ and $0.6 \times (160 - 120)$ oe
	(*their* 30×65 + *their* 70×90 + *their* 40×105 + *their* 36×115 + *their* 24×140) ÷ 200	**M3**	**M1** for midpoints soi **M1** for Σfx, x in interval or boundary of interval **M1 dep on second M1** for ÷ 200
	99.75	**A1**	
4(a)(i)	(2, 7)	**2**	**B1** for each coordinate